Thoughts And Rambles On Probability And Statistics

Problem Set 8: Central Limit Theorems

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These are the end of chapter problems for Chapter 8 of the notes. Equation numbers refer to the full notes.

1. Let $X_\lambda \sim \text{Poisson}(\lambda)$. Using characteristic functions, show that $(X_\lambda - \lambda)/\sqrt{\lambda} \rightsquigarrow N(0,1)$ as $\lambda \to \infty$. Why is this a central limit theorem even though there's no sum in sight?

2. Let $Q_n \sim \chi^2_n$. Show that $(Q_n - n)/\sqrt{2n} \rightsquigarrow N(0,1)$, and use the cumulants to find the skewness of Q_n. Use this to explain the shapes in figure 5.2.

3. A fair coin is flipped 10,000 times. Use the central limit theorem (with a continuity correction) to approximate the probability of getting between 4,900 and 5,100 heads, inclusive. How many flips would you need for the fraction of heads to be within 0.001 of 1/2 with probability 0.99?

4. Let $X_1, X_2, \ldots$ be i.i.d. uniform on $[-1,1]$. Find the exact characteristic function of Z_n, expand $\log \varphi_{Z_n}(t)$ to order t^4, and show that the first correction to the normal approximation is of order $1/n$ rather than $1/\sqrt{n}$.

5. Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and variance σ^2. Find the limiting distribution of $\sqrt{n}(\bar{X}_n^2 - \mu^2)$ when $\mu \neq 0$. When $\mu = 0$, show that $n\bar{X}_n^2/\sigma^2 \rightsquigarrow \chi^2_1$ instead.

6. Let $X_1, X_2, \ldots$ be independent with $\mathbb{P}(X_k = k) = \mathbb{P}(X_k = -k) = 1/2$. Find s_n^2, and check whether the Lindeberg condition holds. Does S_n/s_n converge to a normal?

7. Let $X_1, X_2, \ldots$ be i.i.d. with density $f(x) = |x|^{-3}$ for $|x| \geq 1$. Show that the variance is infinite, but that $S_n/\sqrt{n \log n} \rightsquigarrow N(0,1)$. (Hint: truncate at $\sqrt{n \log n}$ and use the Lindeberg-Feller theorem on the truncated variables. This is the borderline case $\alpha = 2$, and the Gaussian survives with a logarithmic correction to the scaling.)

8. (Computational) Build a Galton board: simulate 10^5 balls each falling through 20 rows of pegs, bouncing left or right with probability 1/2 at each peg, and plot the histogram of final positions against the de Moivre-Laplace approximation. Then make the board biased (probability 0.8 of bouncing right) and see how the agreement changes with the number of rows.